

2.5 NEX (NUMBER OF EXCITATIONS) & AVERAGES

THE POWER OF REPEATING THE SIGNAL

NEX (Number of Excitations) or Averages is one of the most effective ways to improve image quality. It improves SNR and reduces random noise — but the trade-off is longer scan time.



KEY CONCEPT

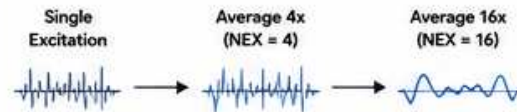
More averages = better SNR and less noise.
But scan time increases linearly.
Use the lowest NEX that gives diagnostic quality.

- ↑ Increase / Improve
- ↓ Decrease / Reduce
- No / Minimal Effect

1 WHAT IS NEX / AVERAGES?

NEX = Number of Excitations
(also called Averages)

The sequence is repeated N times and the signals are averaged to form the final image.



How it Works

Signal is consistent from repetition to repetition.
Random noise is different each time.
When averaged, noise cancels out and signal is reinforced.

2 EFFECT OF NEX ON IMAGE QUALITY

NEX (Averages)	Relative SNR*	Noise Level	Image Appearance
1	1x	High	Noisy, grainy
2	1.4x	↓ ~30%	Slightly less noise
4	2x	↓ ~50%	Good improvement
8	2.8x	↓ ~65%	Less noise, more detail
16	4x	↓ ~75%	Very clean image
32	5.7x	↓ ~82%	Minimal additional improvement

SNR improves with the square root of NEX:
 $SNR \propto \sqrt{NEX}$ (Diminishing returns at higher NEX)

3 EFFECT OF NEX ON SCAN TIME

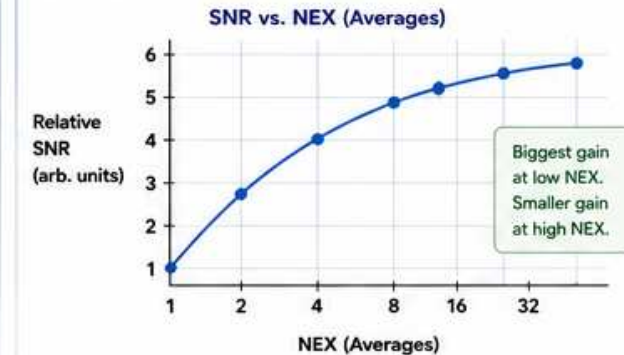
Scan Time $\propto$ NEX

NEX	Relative Scan Time	Scan Time Example (Base = 2:00 min)
1	1x	2:00 min
2	2x	4:00 min
4	4x	8:00 min
8	8x	16:00 min
16	16x	32:00 min



Doubling NEX = doubling scan time

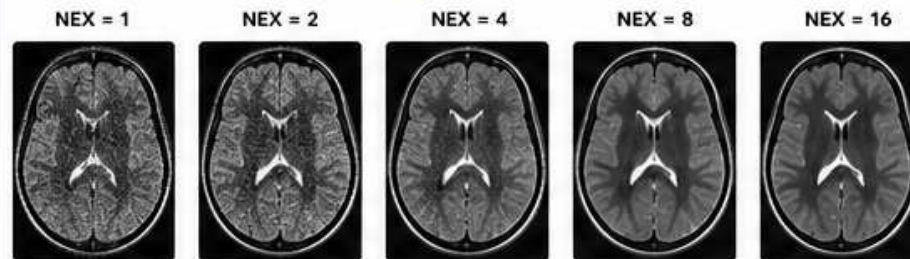
4 NEX & SNR RELATIONSHIP



$SNR \approx \sqrt{NEX} \rightarrow$ Diminishing returns

5 VISUAL EXAMPLES (SAME PATIENT, SAME PARAMETERS EXCEPT NEX)

T2-Weighted Axial Brain



Noisier – Less Detail

Cleaner – More Detail

T2-Weighted Sagittal Spine

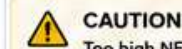


Noisier – Less Detail

Cleaner – More Detail

6 WHEN TO USE HIGHER NEX?

- ✓ Small anatomy (e.g., inner ear, wrist, ankle)
- ✓ Low signal areas or challenging anatomy
- ✓ Higher field strength (3T and above)
- ✓ When motion is minimal
- ✓ MRA or flow studies
- ✓ When highest image quality is needed



CAUTION

Too high NEX can:

- Increase patient time in scanner (motion risk)
- Reduce patient throughput
- Provide minimal additional benefit

7 NEX VS. OTHER PARAMETERS

Parameter	Effect on SNR	Effect on Scan Time	Other Effects
NEX / Averages	↑↑ Significant	↑ Linear increase	Reduces random noise
Bandwidth	↓	—	Affects noise & artifacts
Matrix	↑ (slight)	↑	Improves spatial resolution
Slice Thickness	↑	↓ (slightly)	More signal with thicker slices
Field Strength (B ₀)	↑↑	—	Higher SNR at higher Tesla
Coil Channels	↑↑	—	Better SNR with more channels

8 PRACTICAL GUIDELINES

- 🎯 Use the lowest NEX that provides diagnostic quality.
- ⚖️ Balance NEX with scan time and patient comfort.
- 👤 Consider other options first (coil, bandwidth, matrix, slice thickness) before increasing NEX.
- 🕒 If SNR is still low after other optimizations, then increase NEX.
- ✅ Always evaluate the final image, not the numbers.

9 COMMON NEX SETTINGS (GENERAL GUIDE)

Exam Type / Application	Typical NEX Range
Routine Brain (2D)	1 – 2
Routine Spine (2D)	2 – 4
Extremity (2D)	2 – 4
Abdomen (2D)	2 – 4
High Resolution / Small FOV	4 – 8
MRA (TOF / Contrast)	1 – 4
3D Sequences	1 – 2

10 ENGINEER CHECKLIST

- ✓ Is the NEX appropriate for the clinical goal?
- ✓ Can SNR be improved by other parameters first?
- ✓ Is scan time still acceptable for the patient?
- ✓ Are we balancing quality, time, and patient comfort?
- ✓ Have we documented the rationale if high NEX is used?



KEY TAKEAWAY

Averages improve SNR by reducing random noise.
Use them wisely — enough to see what matters,
not more than what's needed.